

Safety Factor

AP Physics 1 · all eight units

Seven rides at Corbin Park, and an inspector who will not take a number nobody can derive.

45	14	4.9	10/12	3	17·27·1
STOPS	CHOICE (31%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT·CONSTRUCT·OPERATE

READING Where this campaign stands

All three equation gaps are mentioned on DERIVE stops, which means all three are almost certainly Phase 0 phantoms — a derivation of $\Sigma F = ma$ is the arithmetic, and the check cannot currently see a DERIVE’s own lines for these keywords. Expect the real gap count to be zero or one. What is real is three **absent** concepts — free-body thinking, what supplies the centripetal force in each case, and work as force times distance — and three select-only ones. So this campaign’s plan is almost entirely about the fun side, which is fortunate, because its place is the best arcade fit in the catalogue.

PROTECTED – DO NOT CONVERT

All ten DERIVE stops stay. They are the course: an inspector who will not take a number nobody can derive is the premise of the game, and the format is the premise rendered. Nothing in any slate below touches one.

NOW The mix as it ships

CHOICE 14	DERIVE 10	BALLPARK 10	SEQUENCE 3	DIAGNOSIS 2	PROTOCOL 1	TRIGGER 1	VERIFY 1	DELEGATE 1	CASEBOOK 1	VALUE 1
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CURRICULUM What the syllabus is not getting

EQUATION THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
$\Sigma F = ma$	what a net force is for	1·DERIVE 3·CHOICE 5·BALLPARK 9·CHOICE 10·DERIVE	verify first
safety factor = capacity ÷ demand	the number this whole job is about	7·DERIVE 26·CHOICE 29·CHOICE 40·CASEBOOK 44·CHOICE	verify first

2 of 2 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 0 are unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM CONCEPT	STOPS
24	Rotational inertia, and that where the mass is matters more than how much	18
29	Resonance and damping: driving an oscillator in step or out of it	27, 35
30	Measuring g from a period, and why it belongs to the place	3, 21

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects. The syllabus also lists 3 concepts no stop touches at all — **Free-body thinking: one object, every force on it, nothing else; What supplies the centripetal force in each case — a rail, a chain, a seat; Work as a force times a distance along it** — which is expected, and is a different conversation from this one.

FUN What this subject and this place can carry

SWEEP STRONG

stop 27 or 35 · SHIP

Concept 29 — resonance and damping, driving an oscillator in step or out of it — is select-only across two stops, and the pirate ship is a driven pendulum. Drag the drive frequency and the amplitude curve appears only where you looked.

SERVES concept 29, off the select-only list, on the ride that is literally the apparatus

NEEDS a peak that is findable and not marked

TRIAL STRONG

the park

Seven rides, a front gate, a workshop, and an inspection round whose order is the answer — cold checks before the ride runs, brakes before the lift hill. This is the format using the whole site as the question.

SERVES inspection order and ride dependency

RISK world work, and gates must stand off each door by $d/2 + 10$

PROBE STRONG

stop 21 · TOWER

Carry an accelerometer round the coaster and take readings station by station: the place where the pattern changes is not the place with the biggest number. "Measuring gravity with a boat" is a CHOICE and concept 30 is select-only.

SERVES concept 30 — measuring g from a period, and why it belongs to the place

NEEDS a break in the pattern that is not the peak

HOLD STRONG

FLUME or TOWER

Hold the flume level, or the drop-tower brake pressure, inside a closing band while load steps arrive. A ride operator's whole job, and the disturbances are the riders.

SERVES work and energy against a real load

SPOT STRONG

a WHEEL or BUMPER stop

An inspection criterion replaced mid-round with nothing announcing it — which is what happens when a bulletin arrives at nine in the morning. The park is full of items that only one criterion wants.

SERVES acceptance criteria and the cost of not noticing

BELT MEDIUM

BUMPER pavilion

Sorting riders against one restraint criterion at the speed a queue moves. Physically exact and thin on physics — one bit, and the bit is a threshold rather than a mechanism.

RISK must be decidable at a glance, or it is dexterity

LOB DEFER

the ARCADE building

There is a literal arcade on this site, and a ball-toss booth is the one place where a withheld launch speed is diegetic rather than dishonest — a midway game is **designed** so you cannot compute it. But AP Physics 1 asks the student to compute projectiles, so the stop would teach the opposite of its unit. Defer, and argue it again if the withheld quantity can be the booth's rigging rather than the physics.

SLATES Three appetites — the campaign is zero-sum

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: `curriculumDelivery.mjs midway --snapshot before.json`, then `--against before.json`.

Minimum honest 3 STOPS · ONE SITTING

35	d12	SHIP	CHOICE → SWEEP	Three tenths of a second three tenths of a second — the ship as a driven pendulum
18	d6	WHEEL	CHOICE → BALLPARK	Where the mass is where the mass is — computed, since rotational inertia is select-only
21	d7	TOWER	CHOICE → PROBE	Measuring gravity with a boat g measured round the site rather than picked

CHOICE 14 → 11 (24%) · moves 17·27·1 → 14·28·3 · effective formats 4.9 → 5.6 · 2 new operate stops

Balanced 6 STOPS · THE RECOMMENDATION

35	d12	SHIP	CHOICE → SWEEP	Three tenths of a second
18	d6	WHEEL	CHOICE → BALLPARK	Where the mass is
21	d7	TOWER	CHOICE → PROBE	Measuring gravity with a boat
3	d1	SHIP	CHOICE → HOLD	60 people and the same 5 seconds 60 people and the same 5 seconds, held rather than described
29	d10	COASTER	CHOICE → SPOT	A margin that moves a margin that moves, under a criterion that also moves
23	d8	BUMPER	CHOICE → TRIGGER	The tenth of a second that matters the tenth of a second, written as a rule before it matters

CHOICE 14 → 8 (18%) · moves 17·27·1 → 11·29·5 · effective formats 4.9 → 6.5 · 4 new operate stops, 2 of them arcade

Ambitious 9 STOPS · ADDS WORLD WORK

35	d12	SHIP	CHOICE → SWEEP	Three tenths of a second
18	d6	WHEEL	CHOICE → BALLPARK	Where the mass is
21	d7	TOWER	CHOICE → PROBE	Measuring gravity with a boat
3	d1	SHIP	CHOICE → HOLD	60 people and the same 5 seconds
29	d10	COASTER	CHOICE → SPOT	A margin that moves
23	d8	BUMPER	CHOICE → TRIGGER	The tenth of a second that matters
26	d9	TOWER	CHOICE → TRIAL	What a safety factor says the inspection round, driven, in the order that is the answer
9	d3	BUMPER	CHOICE → BELT	100 collisions an hour the restraint criterion at queue speed
42	d14	FLUME	CHOICE → CLOUD	Four adults, and why not five four adults and why not five — a distribution against a limit

CHOICE 14 → 5 (11%) · moves 17·27·1 → 8·30·7 · effective formats 4.9 → 7.4 · 6 new operate stops, 4 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline curriculum-debt.json. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent. 2. Snapshot. 3. Edit the book, never the generated content — tools/import-book.mjs books/midway.yml midway --verify. 4. npm run traps: every instrument's trap must fire. 5. npm run check midway and npm run drive midway, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. 6. --against the snapshot. A changed format is reported; a changed objective fails.

Counts computed from `themes/midway/content/` via `engine/dev/curriculumDelivery.mjs`; concept and equation lists authored in `tools/syllabus.js`. Format verdicts and slates are editorial. Run it with `THEME=midway npm run dev`.